

Orientational Uncertainty and Relational Octaves in the Mersenne Spectrum

A technical note complementary to the geometric-relational framework of the proton charge radius

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Abstract

This note formalizes two arithmetic structures associated with the spectrum

$$M_\varepsilon = 2^\varepsilon - 1,$$

used in the main geometric-relational framework of the proton charge radius. The first structure is *orientational uncertainty*: once it is postulated that each copy of a repeated prime carries an internal binary orientation, a factor p^α with $\alpha \geq 2$ contains exactly $\alpha - 1$ independent relational bits. The second structure is the *relational octave*

$$M_{2k} = M_k(2^k + 1),$$

with coprime factors. This identity shows that every even level contains, in an arithmetically exact way, an inherited copy of level k and a complementary quotient.

The central result of the note is deliberately conditional: the arithmetic is formal; the harmonic reading depends on the interpretative postulates of the framework. Under this reading, even levels may be understood as relational closures of previous levels, and the singularity of $\varepsilon = 6$ is characterized as the first and only case of pure repetition for $\varepsilon \leq 30$. The values with positive orientational uncertainty in that range are:

$$\varepsilon = 6, 12, 18, 20, 21, 24, 30.$$

1 Logical status of the note

The purpose of this note is not to introduce new particles, nor to extend the framework to astrophysical, chemical, or cosmological scales. Its aim is more modest: to isolate two arithmetic patterns in the Mersenne spectrum and to specify which part of those patterns is demonstrable and which part belongs to the physical reading of the program.

We shall use three labels:

Formal. A direct mathematical consequence of definitions, factorizations, or arithmetic identities.

Harmonic. An internal interpretation of the framework. It is consistent with its postulates, but it does not follow from bare arithmetic alone.

Programmatic.

A direction for future work. It is not used as a premise for proving the results of this note.

Warning 1 (Separation between arithmetic and interpretation). *The identity $M_{2k} = M_k(2^k + 1)$ is an elementary theorem. The claim that an even level represents a relation between previous closures is a harmonic reading. Likewise, the count of $\alpha - 1$ bits is formal once a binary orientation per prime copy has been introduced, but that orientation is a postulate of the framework, not a consequence of factorization by itself.*

2 Minimal background

In the main paper, the harmonic depth ε determines the projected quadratic fraction

$$r_V = 2^{-\varepsilon},$$

and the confined signature is written as

$$\frac{\omega_m^2}{\omega_V^2} = 2^\varepsilon - 1 = M_\varepsilon.$$

The first relevant terms are

$$M_1 = 1, \quad M_2 = 3, \quad M_3 = 7, \quad M_4 = 15 = 3 \cdot 5, \quad M_5 = 31, \quad M_6 = 63 = 3^2 \cdot 7.$$

In the main framework, the low-order assignments are

$$\varepsilon = 2 \leftrightarrow \text{electron}, \quad \varepsilon = 3 \leftrightarrow \text{free neutron}, \quad \varepsilon = 4 \leftrightarrow \text{proton}.$$

This note does not require assigning elementary meaning to levels $\varepsilon \geq 5$.

Remark 1 (Depth convention). *In this note, ε always denotes the quadratic harmonic depth of the closure:*

$$r_V = \frac{\omega_V^2}{\omega_I^2} = 2^{-\varepsilon}, \quad \omega_V = \omega_I 2^{-\varepsilon/2}.$$

With this convention, the Mersenne signature is fixed by

$$M_\varepsilon = \frac{\omega_m^2}{\omega_V^2} = 2^\varepsilon - 1.$$

Therefore, ε is the index to be preserved when comparing this note with the main document on the proton charge radius.

3 Orientational uncertainty

3.1 Idea

Suppose that

$$M_\varepsilon = \prod_{i=1}^s p_i^{\alpha_i}$$

is the prime factorization of M_ε . When all exponents α_i are equal to 1, each prime appears only once. In that case there is no internal ambiguity between copies of the same mode.

The situation changes when a factor p^α with $\alpha \geq 2$ appears. In the harmonic reading, this means that the same prime mode appears repeatedly. If each copy carries a binary orientation, then only the relative orientations between copies matter, not the absolute sign of all of them at once.

3.2 Orientational postulate

Postulate 1 (Binary orientation per prime copy). *For each occurrence of a prime p_i in the factorization of M_ε , introduce a variable*

$$\sigma_i^{(j)} \in \{+1, -1\}, \quad j = 1, \dots, \alpha_i.$$

Two assignments that differ by simultaneous inversion of all copies of the same prime are considered physically equivalent:

$$\sigma_i^{(j)} \mapsto -\sigma_i^{(j)} \quad \text{for all } j.$$

Definition 1 (Orientational relations). *For a prime p_i with multiplicity $\alpha_i \geq 2$, define the internal relations*

$$\eta_i^{(j,k)} = \sigma_i^{(j)} \sigma_i^{(k)} \in \{+1, -1\}, \quad 1 \leq j < k \leq \alpha_i.$$

These relations are invariant under global inversion of all copies of p_i .

Theorem 1 (Counting orientational degrees). *Under Postulate 1, a factor $p_i^{\alpha_i}$ contributes exactly $\alpha_i - 1$ independent orientational bits. Therefore,*

$$I_{\text{or}}(\varepsilon) = \sum_{i=1}^s (\alpha_i - 1).$$

Proof. For a fixed prime p_i with multiplicity α_i , there are 2^{α_i} sign assignments. Simultaneous inversion of all signs does not change the internal relations $\eta_i^{(j,k)}$, so the distinguishable configurations are

$$\frac{2^{\alpha_i}}{2} = 2^{\alpha_i-1}.$$

The information required to distinguish them is

$$\log_2(2^{\alpha_i-1}) = \alpha_i - 1$$

bits. Since the choices associated with distinct primes are independent in this combinatorial model, the contributions add. $\square$

Corollary 1. *If M_ε is prime, then $I_{\text{or}}(\varepsilon) = 0$. If M_ε is composite but square-free, then $I_{\text{or}}(\varepsilon) = 0$ as well. Orientational uncertainty appears exactly when M_ε is not square-free.*

4 Low-order cases

Table 1: First values of M_ε and their orientational uncertainty

ε	M_ε	Factorization	$I_{\text{or}}(\varepsilon)$
1	1	unit	0
2	3	3	0
3	7	7	0
4	15	$3 \cdot 5$	0
5	31	31	0
6	63	$3^2 \cdot 7$	1
7	127	127	0
8	255	$3 \cdot 5 \cdot 17$	0
9	511	$7 \cdot 73$	0
10	1023	$3 \cdot 11 \cdot 31$	0
11	2047	$23 \cdot 89$	0
12	4095	$3^2 \cdot 5 \cdot 7 \cdot 13$	1

4.1 Pure repetition

Definition 2 (Pure repetition up to a level). *A level ε with $I_{\text{or}}(\varepsilon) > 0$ is said to exhibit pure repetition if all primes appearing in M_ε had already appeared in some M_j with $1 < j < \varepsilon$.*

Proposition 1 (Singularity of $\varepsilon = 6$ up to $\varepsilon = 30$). *For $1 \leq \varepsilon \leq 30$, the values with positive orientational uncertainty are*

$$\varepsilon = 6, 12, 18, 20, 21, 24, 30.$$

Among them, only $\varepsilon = 6$ is a case of pure repetition.

Proof. The claim is verified by direct factorization of the numbers $2^\varepsilon - 1$ in the indicated range. The cases with repeated exponents are:

ε	M_ε	factorization	$I_{\text{or}}(\varepsilon)$
6	63	$3^2 \cdot 7$	1
12	4095	$3^2 \cdot 5 \cdot 7 \cdot 13$	1
18	262143	$3^3 \cdot 7 \cdot 19 \cdot 73$	2
20	1048575	$3 \cdot 5^2 \cdot 11 \cdot 31 \cdot 41$	1
21	2097151	$7^2 \cdot 127 \cdot 337$	1
24	16777215	$3^2 \cdot 5 \cdot 7 \cdot 13 \cdot 17 \cdot 241$	1
30	1073741823	$3^2 \cdot 7 \cdot 11 \cdot 31 \cdot 151 \cdot 331$	1.

For $\varepsilon = 6$, the primes 3 and 7 had already appeared in M_2 and M_3 . In every other case, at least one new prime appears relative to all previous levels: 13 at $\varepsilon = 12$, 19 and 73 at $\varepsilon = 18$, 41 at $\varepsilon = 20$, 337 at $\varepsilon = 21$, 241 at $\varepsilon = 24$, and 331 at $\varepsilon = 30$. Therefore, only $\varepsilon = 6$ satisfies the condition of pure repetition in this range. $\square$

Remark 2 (Arithmetic check). *The value $\varepsilon = 28$ does not belong to the list of levels with positive orientational uncertainty, since*

$$M_{28} = 2^{28} - 1 = 3 \cdot 5 \cdot 29 \cdot 43 \cdot 113 \cdot 127,$$

is square-free and therefore has $I_{\text{or}}(28) = 0$. By contrast, $\varepsilon = 21$ does belong to the list, since

$$M_{21} = 7^2 \cdot 127 \cdot 337,$$

so $I_{\text{or}}(21) = 1$.

5 Relational octaves

5.1 Fundamental identity

Theorem 2 (Octave factorization). *For every $k \geq 1$,*

$$\boxed{M_{2k} = M_k Q_k, \quad Q_k = 2^k + 1.}$$

Moreover,

$$\gcd(M_k, Q_k) = 1.$$

Proof. The identity follows from the difference of squares:

$$M_{2k} = 2^{2k} - 1 = (2^k - 1)(2^k + 1) = M_k Q_k.$$

If $d = \gcd(M_k, Q_k)$, then

$$d \mid (Q_k - M_k) = 2.$$

But $M_k = 2^k - 1$ is odd for $k \geq 1$, so d is also odd. The only odd divisor of 2 is 1. Hence $\gcd(M_k, Q_k) = 1$. $\square$

5.2 Harmonic reading of the octave

The formal identity says that the level $2k$ contains two coprime factors:

$$M_{2k} = \underbrace{M_k}_{\text{inherited signature}} \underbrace{(2^k + 1)}_{\text{complementary quotient}}.$$

The proposed harmonic reading is that the level $2k$ may be interpreted as a relation between a previous closure M_k and complementary information Q_k . This interpretation does not claim, by itself, that every even level is physically elementary, nor that every quotient Q_k corresponds to a particle.

Table 2: First relational octaves				
k	M_k	$Q_k = 2^k + 1$	M_{2k}	Harmonic reading
1	1	3	3	First non-trivial closure of the unit mode.
2	3	5	15	The level $\varepsilon = 4$ contains the signature M_2 plus a coprime complement.
3	7	$9 = 3^2$	63	The level $\varepsilon = 6$ combines the signature M_3 with a quadratic repetition of the prime 3.
4	15	17	255	The level $\varepsilon = 8$ inherits the composite closure 3·5 and adds a new prime.
5	31	$33 = 3 \cdot 11$	1023	The level $\varepsilon = 10$ relates a prime signature to a composite complement.
6	63	$65 = 5 \cdot 13$	4095	The level $\varepsilon = 12$ carries the repetition of $\varepsilon = 6$ and adds 13.

6 Compatibility with the orientation σ of the main framework

In the main paper, the double cover introduces an internal orientation

$$\sigma \in \{+1, -1\}$$

associated with the minimal spinorial structure. The orientational uncertainty defined here also produces binary choices, especially when a quadratic factor p^2 appears.

The similarity is formal, but it must not be confused with conceptual identity. For example, the proton in the main framework corresponds to $\varepsilon = 4$ and

$$M_4 = 15 = 3 \cdot 5,$$

so

$$I_{\text{or}}(4) = 0.$$

Nevertheless, the proton does possess the orientation σ of the double cover. Therefore, the spinorial orientation of the main framework does not reduce to prime repetition. The correct relation is weaker:

$$\text{prime repetition} \implies \text{possible additional orientational bit},$$

but

$$\text{fundamental spinorial bit} \not\Rightarrow \text{prime repetition}.$$

7 Programmatic directions

The following lines of work are natural, but they are not part of the results proved in this note:

1. **Perpendicular projection operator.** Formalize when an orientational uncertainty cannot be absorbed inside the coherent geometry $\mathcal{G}_\varepsilon$ and must be projected as an external degree of freedom.
2. **Generalization to higher multiples.** For $q > 2$,

$$M_{qk} = M_k \sum_{j=0}^{q-1} 2^{jk}.$$

The quotient

$$R_{q,k} = \sum_{j=0}^{q-1} 2^{jk}$$

could classify higher-order relations.

3. **Statistical growth of $I_{\text{or}}(\varepsilon)$.** Study how often M_ε ceases to be square-free and the distribution of prime multiplicities in the spectrum.
4. **Physical admissibility criterion.** Separate arithmetically interesting levels from physically admissible levels. Arithmetic suggests candidates; the physical framework must impose additional filters.

8 Conclusions

This note establishes three main results:

1. Orientational uncertainty is not an automatic property of Mersenne numbers: it is a formal quantity *conditioned* on the postulate of binary orientation per prime copy.
2. Under that postulate, the formula

$$I_{\text{or}}(\varepsilon) = \sum_i (\alpha_i - 1)$$

measures exactly the number of independent orientational bits associated with the prime multiplicities of M_ε .

3. The identity

$$M_{2k} = M_k(2^k + 1)$$

provides a structure of coprime octaves. The harmonic reading interprets those even levels as relations between previous closures and complementary quotients, but that reading must remain separate from the arithmetic theorem.

The case $\varepsilon = 6$ is highlighted without exaggeration: not because it is the only level with orientational uncertainty, but because it is the first and only case of pure repetition up to $\varepsilon = 30$. In it,

$$M_6 = 63 = 3^2 \cdot 7,$$

and the single additional bit comes from the repetition of the prime 3.

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References

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